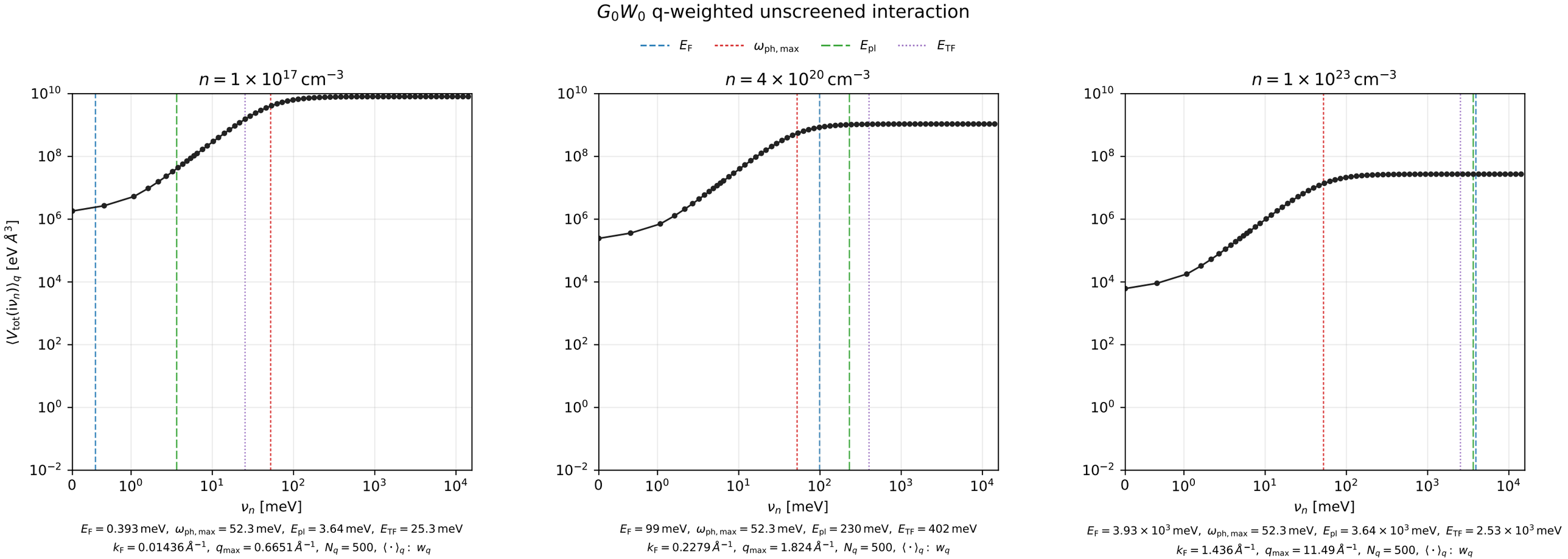
G_0W_0 q-weighted unscreened interaction
E_F
$\omega_{ph,max}$
E_{pl}
E_{TF}
$n = 1 \times 10^{17}\,cm^{-3}$
$n = 4 \times 10^{20}\,cm^{-3}$
$n = 1 \times 10^{23}\,cm^{-3}$
$\langle V_{tot}(i\nu_n)\rangle_q$ [eV Å^3]
ν_n [meV]
$E_F = 0.393$ meV, $\omega_{ph,max} = 52.3$ meV, $E_{pl} = 3.64$ meV, $E_{TF} = 25.3$ meV
$k_F = 0.01436$ Å^{-1}, $q_{max} = 0.6651$ Å^{-1}, $N_q = 500$, $\langle\cdot\rangle_q$: w_q
$E_F = 99$ meV, $\omega_{ph,max} = 52.3$ meV, $E_{pl} = 230$ meV, $E_{TF} = 402$ meV
$k_F = 0.2279$ Å^{-1}, $q_{max} = 1.824$ Å^{-1}, $N_q = 500$, $\langle\cdot\rangle_q$: w_q
$E_F = 3.93 \times 10^3$ meV, $\omega_{ph,max} = 52.3$ meV, $E_{pl} = 3.64 \times 10^3$ meV, $E_{TF} = 2.53 \times 10^3$ meV
$k_F = 1.436$ Å^{-1}, $q_{max} = 11.49$ Å^{-1}, $N_q = 500$, $\langle\cdot\rangle_q$: w_q